

EE-482: Network Security and Cryptography

LECTURE NO 17: *Internet Security Protocols*

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Organization of Lecture

1. Basic Concepts
2. Secure Socket Layer (SSL) Protocol
3. Secure Hyper Text Transfer Protocol (SHTTP)
4. Time Stamping Protocol (TSP)

1. Basic Concepts (1/4)

↳ World Wide Web (www) is a client/server application.

↳ client: Web browser

server: Web server

↳ Hyper Text Transfer Protocol (HTTP) is used for communication b/w browser & server.

→ Request - Response protocol

↳ Web page is written in Hyper Text Mark up Language (HTML).

1. Basic Concepts (2/4)

↳ Static Web pages:

- Browser sends HTTP request
- Server sends HTTP response
- Server locates the page & sends back page without any additional processing
- Static web pages are suitable for contents that do not change often.

↳ Dynamic Web pages:

- contents change often
- more complex than static pages
- Server side runs an application program for updating latest changes

Example: Static page: Country's Home Page, Organizations Information etc
Dynamic page: Latest stock prices

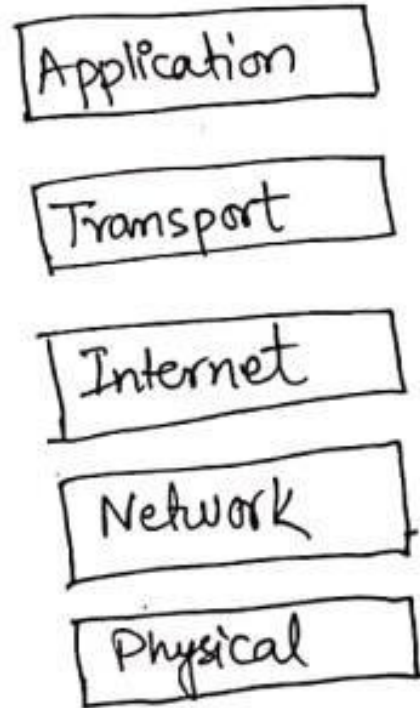
1. Basic Concepts (3/4)

Protocols and TCP/IP

- ↳ Internet is made up of computers and Networks that have different hardware as well as software.
- ↳ Need a universal translator to facilitate communication between computers.
- ↳ Transmission Control Protocol/Internet Protocol (TCP/IP):
 - Combination of many protocols that facilitate the communication between computers over the internet.

1. Basic Concepts (4/4)

↳ TCP/IP protocol has layered organization consisting of 5 layers.



↳ layers add their own header to data packet

↳ Each layer performs a specific pre-determined task.

2. Secure Socket Layer (SSL) Protocol (1/19)

1) Secure Socket Layer (SSL) Protocol

- ↳ An Internet protocol for secure exchange of Information b/w a web browser (client) and web server.
- ↳ Provides authentication and confidentiality
- ↳ Currently 3 versions:
 - version 2
 - version 3
 - version 3.1

2. Secure Socket Layer (SSL) Protocol (2/19)

SSL in TCP/IP Protocol Suite

- ↳ located between application layer and the transport layer.
- ↳ Application layer of sending computer prepares data to be sent.
- ↳ Application layer data is passed to the SSL layer which performs encryption on the data & also adds its own encryption info. header called SSL Header (SH) to encrypted data.
- ↳ Output of SSL layer becomes input for transport layer; adds own header & passes to Internet layer & so on.
- ↳ At receiver end; SSL layer removes header (SH) & performs decryption.
- ↳ SSL only encrypts the application layer data.

2. Secure Socket Layer (SSL) Protocol (3/19)

How SSL works?

3 sub-protocols:

- (i) Handshake protocol
- (ii) Record protocol
- (iii) Alert protocol

i) Handshake Protocol:

↳ used by client and server to communicate.

↳ consists of a series of messages; each with following format

TYPE	LENGTH	CONTENT
1 byte	3 bytes	1 or more bytes

2. Secure Socket Layer (SSL) Protocol (4/19)

TYPE: Field indicates one of the 10 possible message types.

hello request
client hello
Server Hello
Certificate
Server key exchange
Certificate request
Server hello done
Certificate verify
Client Key exchange
Finished.

LENGTH: Field indicates length of message in bytes

CONTENT: Parameters associated with message.

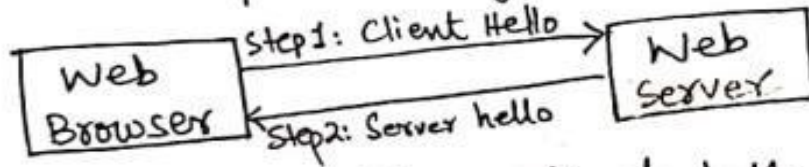
Handshake protocol is made up of 4 phases:

- 1) Establish security capabilities
- 2) Server authentication & key exchange
- 3) Client authentication & key exchange
- 4) Finish.

2. Secure Socket Layer (SSL) Protocol (5/19)

Phase 1: Establish Security Capabilities:

→ Consists of 2 messages: client hello & server hello



→ Process starts with a client hello message.

→ Client hello contains following parameters:

↳ Version: of SSL (can be 2, 3, 3.1)

↳ Random: → 32-bit date time field that identifies current system date & time.

↳ 28-byte random no. generated by random no. generator.

↳ Session id: → Non-Zero value: already connection exists.
→ Zero value: client wishes to create new connection.

↳ Cipher suite: Contains list of cryptographic algorithms supported by client.

↳ Compression method: list of compression algorithms supported by client.

2. Secure Socket Layer (SSL) Protocol (6/19)

↳ client sends client hello message to server and waits for Server response.

↳ Server sends a server hello containing following fields:

→ version: can be 2, 3 or 3.1

→ Random: Same as that of client hello except the 28-byte random no. generated is completely independent of client's random value.

→ Sessionid:
↳ if non-zero, server uses same value.
↳ otherwise server creates new session id and puts in this field.

→ Cipher suites: Contains single cipher suite selected from the list sent earlier by client.

→ Compression method.

↳ Contains single compression method selected from the list sent earlier by client.

2. Secure Socket Layer (SSL) Protocol (7/19)

Phase 2: Server authentication & key exchange

- ↳ Initiated by the Server & server is sole sender of all messages.
- ↳ Client is sole recipient of all messages
- ↳ Contains 4 steps



2. Secure Socket Layer (SSL) Protocol (8/19)

Step 1: Certificate: Server sends its digital certificate to help client to authenticate it using server's public key.

Step 2: Server key exchange } OPTIONAL: Server sends public key only if server doesn't send its digital certificate in step 1.

Step 3: Certificate request } Server can request for client's digital certificate. OPTIONAL

Step 4: Server hello done } Indicate to client that server's portion of hello message is complete.

2. Secure Socket Layer (SSL) Protocol (9/19)

Phase 3: Client Authentication & Key Exchange

↳ initiated by client & client is sole sender of all messages.

↳ 3 steps

Step 1: Certificate: **OPTIONAL**. Performed only if server has requested for client's digital certificate

If client doesn't have a certificate; it sends a **NO CERTIFICATE** message.

Step 2: Server key exchange: Client sends information related to symmetric key that both parties will use in this session.

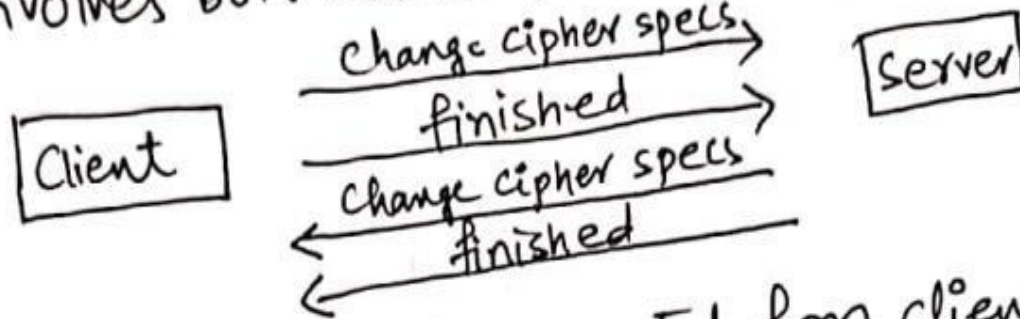
Step 3: Certificate Verify: Only if server has demanded client authentication.

↳ Client needs to prove to server that it is the correct & authorized holder of private key corresponding to certificate.

2. Secure Socket Layer (SSL) Protocol (10/19)

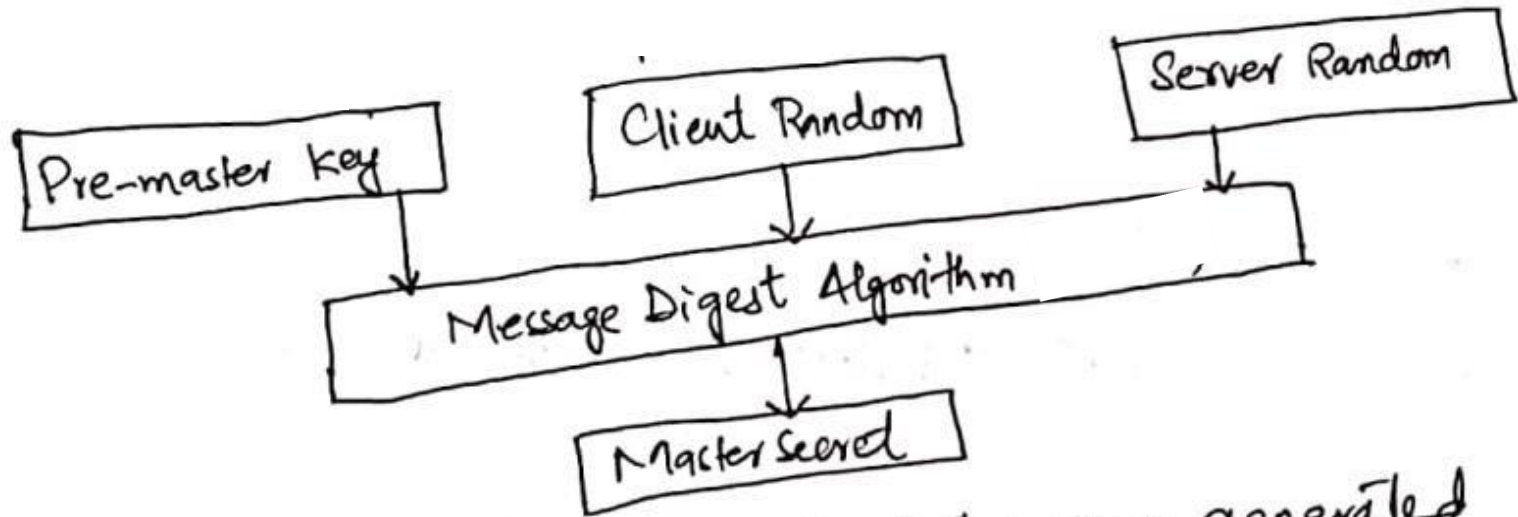
Phase 4: Finish

↳ involves both client & server

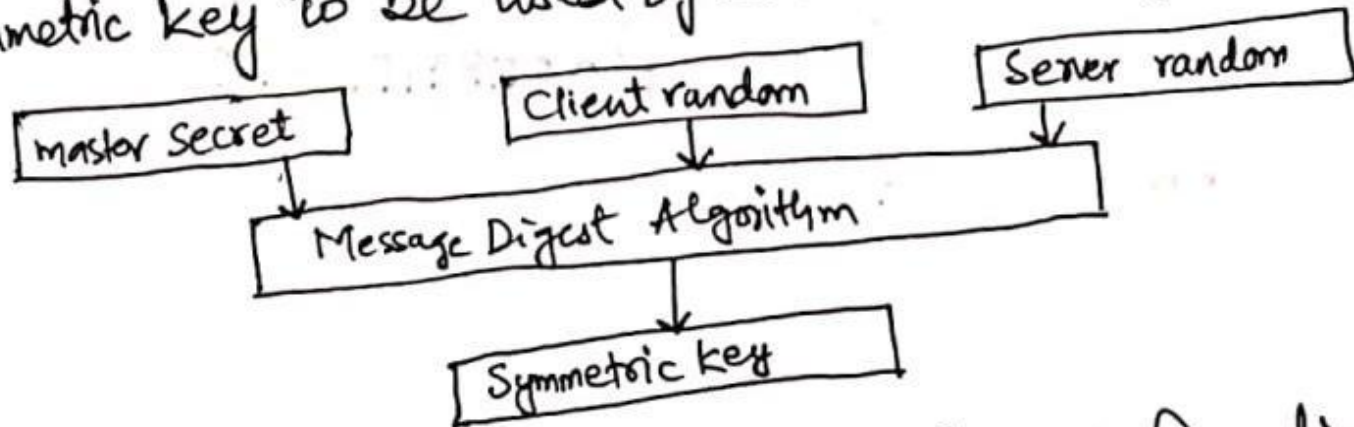


- ↳ First 2 messages are generated from client
- ↳ In phase 3; client creates a pre-master secret key & sends it to server (in encrypted form).
- ↳ Based on pre-master secret key both client & server create a master secret.
- ↳ Master secret is used to generate keys & secrets for encryption & MAC computations.

2. Secure Socket Layer (SSL) Protocol (11/19)



↳ Symmetric key to be used by client & server generated



↳ After this the change cipher specs is a confirmation from client that all is well from its end.

2. Secure Socket Layer (SSL) Protocol (12/19)

ii) Record Protocol:

↳ comes into play after successful handshake completed.

↳ Client & Server have authenticated each other & have decided which algorithms to use for secure info. exchange.

↳ Provides 2 Services:

Confidentiality: Achieved by using secret key defined by handshake protocol.

Integrity: Handshake protocol also defines a shared secret (MAC) that is used for assuring message integrity.

↳ SSL record protocol takes an application message as input.

↳ 5 steps:

2. Secure Socket Layer (SSL) Protocol (13/19)

↳ 5 steps:

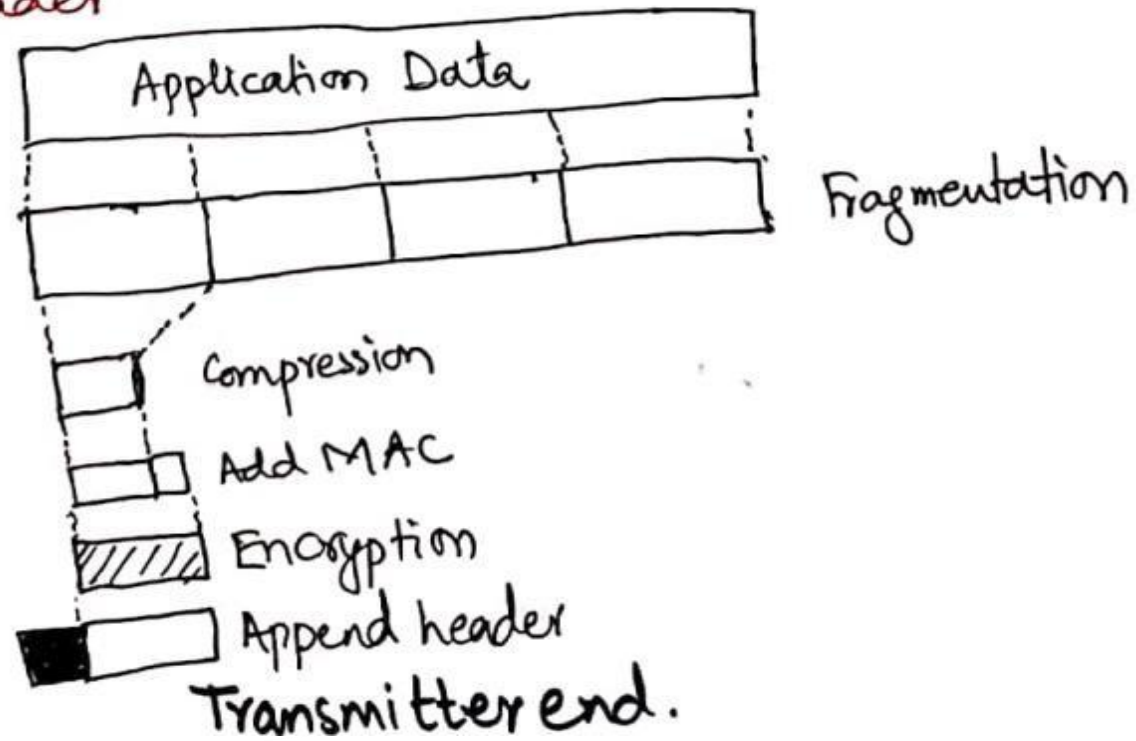
Fragmentation

Compression

Addition of MAC

Encryption

Append header



2. Secure Socket Layer (SSL) Protocol (14/19)

Step 1: Fragmentation

Original application message is broken into blocks each less than or equal to 16 kb.

Step 2: Compression

The compression process must not result into loss of original data → use loss less compression mechanism.

Step 3: Addition of MAC

Using shared key established in handshake protocol, Message Authentication Code (MAC) for each block is calculated.

2. Secure Socket Layer (SSL) Protocol (15/19)

Step 4: Encryption

Using symmetric key established in handshake protocol output of step 3 is encrypted.
Encryption should not increase the overall size of block by more than 1024 bytes.

Step 5: Append header

→ Header added to encrypted block.

→ Header contains

Content type (8 bits): Specifies protocol used. e.g. handshake

Major version (8 bits): major version of SSL protocol
e.g. 3.1

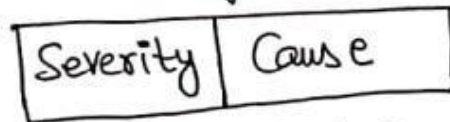
Minor version (8 bits): minor version of SSL protocol
e.g. 3.1

Compressed length (16 bits): Specifies length in bytes of original plain text block (or compressed block)

2. Secure Socket Layer (SSL) Protocol (16/19)

iii) Alert Protocol:

- ↳ When either client or server detects an error, detecting party sends an alert message to other party.
- ↳ If error is fatal, both parties immediately close SSL connection.
- ↳ Both parties also destroy session identifier e.g.: Secrets & keys associated with this connection.
- ↳ Each alert message consists of 2 bytes



Byte 1

Byte 2

Specifies type of error

Specifies actual error

- 1: Warning
- 2: Error Fatal

2. Secure Socket Layer (SSL) Protocol (17/19)

Fatal Alerts

<u>Alert</u>	<u>Description</u>
→ Unexpected message	An inappropriate message was received.
→ Bad record MAC	Message received without correct MAC.
→ Decompression failure	Input to decompression function is inappropriate
→ Handshake failure	Sender was unable to negotiate an acceptable set of security parameters
→ illegal parameters	A field in handshake message was out of range or inconsistent with other field.

2. Secure Socket Layer (SSL) Protocol (18/19)

Receiver's End:

- Header of each block removed
- Block decrypted
- Verified
- Decompressed
- Reassembled into application messages.

2. Secure Socket Layer (SSL) Protocol (19/ 19)

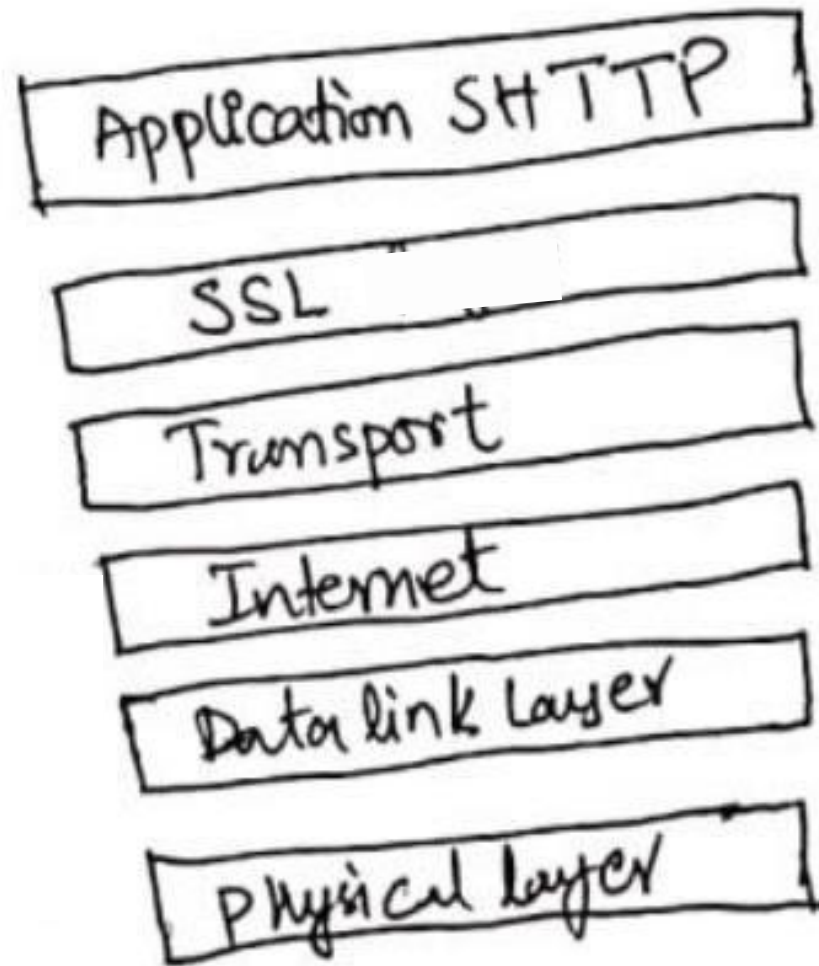
* Closing & Resuming SSL Connections:

- ↳ Client & server must inform each other that communication is ending.
- ↳ Each party sends a close notify alert to other party.
- ↳ If an SSL connection ends without close notify from either party, it cannot be resumed.
- ↳ Handshake protocol in SSL is quite complex as it uses asymmetric key cryptography.
- ↳ If desired, client & server can decide to reuse or resume an earlier SSL connection rather than creating a fresh one.
- ↳ Both parties should agree on the reuse.
- ↳ SSL specification: Any SSL connection should not be reused after 24 hours in any case.

3. Secure Hyper Text Transfer Protocol (SHTTP) [1/2]

- ↳ Security mechanisms defined for protecting internet traffic
- ↳ Includes internet-based transactions.
- ↳ Services offered by SHTTP are quite similar to those of SSL; however SSL is more popular
- ↳ SHTTP works at application layer & provides authentication and encryption of HTTP traffic b/w client and server

3. Secure Hyper Text Transfer Protocol (SHTTP) [2/2]



4. Time Stamping Protocol (TSP) [1/2]

- ↳ Provides proof that a certain piece of data existed at a particular time.
- ↳ This service is provided by an authority called time stamping authority (TSA).
- ↳ TSA acts like a trusted third party.
- ↳ TSP is a request-response protocol.

4. Time Stamping Protocol (TSP) [2/2]

Step 1: MD calculation

→ client requesting a time stamp because it calculates a MD of original message which needs time stamp from TSA.

Step 2: Time Stamping request

→ client sends MD calculated in Step 1 to TSA.
→ This is called a Time Stamping Request

Step 3: Time Stamping Response

→ TSA might decide to grant or reject the time stamp request

→ If decides to accept; TSA signs the client's request together with the time stamp by the TSA's private key.

References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *“Cryptography and Network Security Principles and Practice”, by William Stallings, 6th Edition, Pearson.*
3. *“Computer Security Principles and Practice”, by William Stallings, Lawrie Brown, 4th Edition, Pearson.*